

AI Study Companion - Full Stack MERN + AI Project Proposal

1. Problem Statement

Students often struggle with large PDF materials and lack interactive tools to summarize, understand, and test their knowledge efficiently.

2. Proposed Solution

AI Study Companion is a full-stack web application that transforms static PDF materials into interactive learning experiences using AI.

- Automatic PDF summarization
- AI-generated quizzes and flashcards
- Context-aware AI chat assistant
- Progress tracking and performance analytics
- Adaptive learning recommendations

3. Core Features

- PDF Upload & Text Extraction
- AI Summary Generation
- Auto-Generated MCQ & Short Answer Questions
- Interactive Quiz System with Scoring
- Explain Like I'm 10 Mode
- AI Chat using RAG (answers based only on uploaded PDF)
- Performance Dashboard & Weakness Detection

4. Technical Architecture

- Frontend: React.js + Bootstrap
- Backend: Node.js + Express
- Database: MongoDB

- Authentication: JWT
- AI Integration: Google Gemini API
- RAG Implementation for contextual answering

5. Key Learning Outcomes

- Full-stack MERN development
- AI API integration & prompt engineering
- PDF processing & text extraction
- Adaptive learning logic implementation
- Secure authentication & data management

6. Future Enhancements

- Collaborative study groups
- Integration with LMS systems
- AI-generated presentation slides from PDF